Tooth-scale Stress Localisation in a μCT-derived *Otodus megalodon* Model: Calibrated Geometry, Numerical Verification and Regional Sensitivity

Eliguli Han
Mathematics and Computer Science, New York University
Faculty advisor in the original project record: Leslie Greengard
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Abstract

A single fossil tooth assigned to *Otodus megalodon* provides a geometric basis for a controlled numerical stress experiment, but its interpretation depends on imaging calibration, the solved domain and attachment assumptions. An audit of 3,050 archived DICOM headers identifies isotropic 18 μm spacing and a legacy reconstruction that forced the through-slice spacing to 16 μm. Native CT segmentation and legacy appearance meshes are therefore distinguished from the recalibrated closed-solid finite-element proxy. Three nested P1 tetrahedral meshes contain 55,852, 271,300 and 963,710 elements. Under a specified 1 MPa crown-face pressure and bilateral fixed root-cap volumes, inverse volume-weighted $Q_{0.95}$ stresses are 10.555, 29.587 and 14.572 MPa in the central and two edge-side regions, 0.533 MPa at the macroscopic apex, 62.811 MPa in the transition band and 261.966 MPa in the root-side band. Five regional $Q_{0.95}$ values change by less than 2% between the last two meshes; the apex changes by 5.89%. Equilibrium, energy, affine-patch, rigid-motion, quantile readback and scaling checks support numerical implementation consistency. Sixteen coarse-mesh sensitivity cases show that the two sides can exchange their stress ranking, while the root-side and transition bands remain the two largest regional summaries in this tested set. The results contradict the archived tip-dominant, low-stress-root ordering for this explicitly defined experiment. They establish conditional regional stress contrasts, without demonstrating individual serration-tip failure, a fossil fracture threshold or an organism-level feeding or ecological mechanism. **Keywords:** fossil tooth; micro-computed tomography; linear elasticity; stress quantile; geometric calibration; numerical verification.

1 Introduction

The broad triangular crown and serrated cutting edges of a tooth assigned to *Otodus megalodon* motivate a mechanical question: where does a specified load concentrate stress within this particular geometric model? Answering that question requires a traceable chain from image coordinates to a finite-element field. A coloured model, a dense surface mesh and a converged algebraic solver each address different parts of that chain. None alone establishes tooth strength or feeding behaviour.

Experiments on extant shark teeth show why contact and loading protocol matter: dynamic tooth–prey tests investigate cutting performance directly [1], while tooth-scale structural models and jaw-scale bite analyses investigate different mechanical systems [2–4]. Serration morphology is documented in comparative extant material [5]. These studies motivate the present mechanical questions but do not calibrate this fossil specimen's strength, attachment or loading history. Tooth mineralisation patterns and dental homologies also require specimen-specific anatomical interpretation [6, 7].

This report reconstructs the specimen-derived numerical study with an explicit evidentiary boundary. The specimen is treated as one isolated tooth; its collection identifier, dental position and mesial/distal orientation are not independently established in the available record. The former anterior/A2 assignment is therefore not used as a numerical input. The two original photographs and a recovered render are shown together in Figure 1. A visible surface defect in a photograph is an observation of appearance; it does not identify when or how damage occurred.



(a) Original photograph a. **(b)** Original photograph b. **(c)** Recovered animation frame.

Figure 1: Specimen photographs recovered losslessly from the original report, beside the first native-resolution frame of the recovered natural-colour rotation. Panel (c) is an illustrative, photograph-referenced render of legacy geometry, not a third photograph or an independent tissue measurement. The photographs lack a documented physical scale; their printed sizes are for comparison of appearance only. No apparent texture boundary is identified here as a segmented enameloid–dentine junction.

The study has three quantitative aims. First, it determines which geometric dimensions and internal-void claims are supported by the archived CT record. Second, it specifies and verifies a fresh pressure experiment whose regional outputs can be regenerated from element stresses and saved masks. Third, it tests how those outputs change under refinement and selected support, load and elastic-parameter variations. The original regional field and six anatomical index sets were unavailable, so the original numerical values cannot be reproduced as the same experiment. Table 9 is a replacement result with explicitly defined geometric regions.

2 Imaging calibration, geometry and image provenance

2.1 DICOM calibration and segmentation

All 3,050 archived header records specify 0.018 mm spacing in both pixel directions, 0.018 mm slice thickness and consistent successive slice positions. Fresh reads of the first, middle and last original DICOM headers agree with that record. Thus the available archive supports isotropic 18 μm sampling, rather than the former manuscript's $(50, 50, 100)$ μm entry. The directory name `DICOM_16um` is not a calibration measurement.

The legacy conversion script explicitly overwrote through-slice spacing with 0.016 mm. For the archived closed-solid finite-element geometry this revision applies

$$(x, y, z)_{\text{calibrated}} = (x, y, (18/16)z)_{\text{legacy}}. \tag{1}$$

This is a calibration correction, not an isotropic rescaling. It cannot reconstruct interior background that the older pipeline filled. Nor does it make an independently simplified finite-element boundary identical to the native CT surface.

A slope/intercept pair alone does not establish calibration to Hounsfield units. Grey values and the threshold are therefore reported in scanner-native units. The selected material contains 967,429,317 voxels, 99.87593% of the thresholded foreground. The reconstruction record excludes 345,466 disconnected material specks, the largest containing 931 voxels. Such threshold-defined objects may include anatomical structures, noise or partial-volume effects; their biological identity is not determined by connected-component counting.

Table 1: Evidence objects and the questions each can support.

Object	Direct use in this report	Evidentiary boundary
Two specimen photographs	Appearance and visible surface features	No scale, damage chronology or tissue classification
DICOM and native binary mask	Spacing, segmented material and threshold-defined voids	No independent histology or HU calibration
Native/optimised CT surfaces	Topology and sampled geometric comparison	Surface topology is not anatomical pore identity
Legacy PBR appearance models	Illustrative front/oblique/reverse appearance	No raw numerical stress field in their colour metadata
Closed-solid proxy and new fields	Defined pressure experiment and regional summaries	Single material; native pores and EDJ not represented
Archived COMSOL views/exports	Historical load and parameter audit	No reproducible original 1 MPa regional experiment

Table 2: Imaging metadata supported by the archived scan.

Quantity	Recorded value and interpretation
Original array order	3,050 slices × 2,028 rows × 1,676 columns (Z, Y, X)
Physical spacing	$(0.018, 0.018, 0.018)$ mm in X, Y, Z
Original DICOM LPS origin	$(4.176, 0.198, 16.818)$ mm
Grey-value transformation	RescaleSlope $= 0.5246951$; RescaleIntercept $= -1000$; RescaleType absent in the spot checks
Segmentation threshold	10,361.669921875 raw scanner units; retained from the archived protocol
Native mask	$746 \times 1,903 \times 2,886$ voxels (X, Y, Z), 18 μm spacing
Mask LPS origin	$(11.232, 1.566, 18.186)$ mm
Foreground selection	Largest global 6-connected material component; no binary closing or interior hole filling in the native reconstruction

The native segmentation is saved losslessly. Its reconstruction record reports no voxel downsampling, binary closing, interior filling or separate smoothing filter. Opposing zero-thickness triangle pairs were cancelled at extraction/seam stages; no synthetic cavity caps were inserted in the native reconstruction. This record differs from the legacy appearance model, which underwent filling and local capping. A low/baseline/high threshold experiment and independent segmentation repeatability study have not been performed for the present revision.

2.2 Three distinct geometric objects

Table 3 separates native CT surfaces, appearance geometry and the mechanical proxy. The native surface has a signed material volume of 5,649.938794 mm^3. Its 803,442 inward-oriented closed components enclose 12.936868 mm^3; these are segmented surface components, not a confirmed count of anatomical pores. The optimised surface retains the recorded component count and Euler characteristic but encloses only 10.990583 mm^3 of closed void space, 15.04% less. Topology retention therefore does not imply exact pore-shape or pore-volume preservation.

The archived bidirectional surface sampling tested 145,352 points. Its 99th-percentile and maximum sampled distances were 3.808 and 8.801 μm. These are sample statistics, not an exhaustive Hausdorff bound. Across seven sampled axial sections, low-density pixel retention was 2.46% in the legacy supplied model, 98.94% in the optimised CT model and 100% in the native reconstruction. These values compare threshold-derived section occupancies; they do not measure whole-volume anatomical accuracy.

The mechanical domain has volume 5,771.937561 mm^3, 2.159% greater than the native signed material volume. The discrepancy includes differences in boundary approximation and internal void treatment; it cannot be assigned solely to pore filling. All mesh-refinement statements below refer to this fixed proxy domain. They are not convergence statements toward a pore-resolving model of the native fossil.

2.3 Topology and curvature checks

A closed triangulated surface requires more than every edge having two incident faces. Vertex links must also have the local manifold structure; for example, two closed surfaces touching only at one vertex satisfy the edge-incidence condition but have a disconnected vertex link. An embedded, physically meaningful boundary additionally requires geometric intersection and orientation checks. The archived native and optimised reports include exhaustive edge incidence and oriented vertex-link checks, with no reported boundary or nonmanifold edges/vertices. They do not include an exhaustive whole-model self-intersection test. Positive signed volume or a clean rendered image cannot supply that missing test.

Robust tetrahedral meshing can accept imperfect triangle soups [8]; a nonwatertight input is therefore not a universal proof that meshing is impossible. It remains a problem of physical-domain definition. Likewise, in a two-pass Taubin filter a positive averaging step tends to shrink the surface and a negative step compensates; the procedure does not guarantee exact volume or notch preservation [9]. No new smoothing is applied to the solved proxy in this revision.

For an interior surface vertex i, the area-normalised discrete Gaussian curvature and cotangent mean-curvature vector are [10]

$$K_i = \frac{2\pi - \sum_{f \ni i} \theta_{if}}{A_i}, \tag{2}$$

$$\mathbf{k}_i = \frac{1}{2A_i} \sum_{j \sim i} (\cot \alpha_{ij} + \cot \beta_{ij})(\mathbf{x}_i - \mathbf{x}_j), \tag{3}$$

$$|H_i| = \tfrac{1}{2}\|\mathbf{k}_i\|, \qquad H_i = \tfrac{1}{2}\mathbf{k}_i \cdot \mathbf{n}_i. \tag{4}$$

Here A_i is mixed Voronoi area and the last sign uses an outward normal, so a convex sphere has positive H. The angle deficit alone is an integrated dimensionless curvature; omitting A_i does not give pointwise K. With coordinates in mm, K has units mm^{-2} and H has units mm^{-1}.

The implementation passes a triangulated 10 mm-radius sphere check: the integrated Gaussian curvature is 12.5663706144, consistent with 4π, and median signed mean curvature is 0.1000000460 mm^{-1}. Doubling all lengths reduces K by four and H by two. Figure 3 is recomputed from authentic CT-derived arrays; it replaces the screenshot with corrected units and common colour limits. Neither matching colours nor these benchmarks establish that every fossil notch survives arbitrary resampling.

2.4 Recovered appearance figures

The natural and reference-colour triptychs supplied with the revision were located as original 2,400×1,320-pixel exports. Their metadata identifies the colours as photograph/paper-image referenced and states explicitly that no raw numerical FEA stress field underlies the reference colours. Figure 4 preserves those original pixels, removes the old display headers through recorded cropping and states their actual role. Reverse-face colour continuation and roughness are illustrative rather than measured. The 32,687,264-face appearance geometry is the legacy model in Table 3, not the recalibrated native reconstruction or the L2 stress domain.

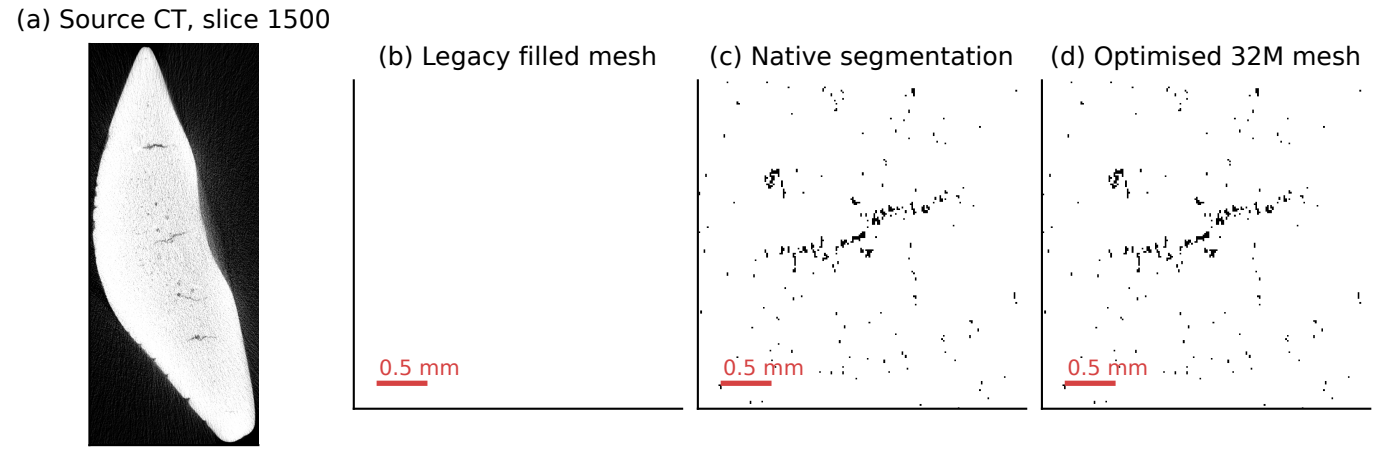


Figure 2: Authentic section data re-rendered from archived CT/mesh comparison arrays. (a) The source CT at slice 1500. (b–d) The same cropped coordinate window in the legacy filled geometry, native threshold segmentation and optimised CT mesh. The red bars are 0.5 mm using the 18 μm pixel spacing. Dark interiors denote threshold-defined background/voids, not histologically identified tissue. The seven-section retention diagnostic and its limits are described in the text.

Table 3: Surface geometry and the mechanical proxy are different numerical objects.

Object	Vertices/nodes	Faces/elements	Interpretation
Native 18 μm CT surface	37,951,699	74,293,846	Surface triangles; internal void boundaries retained
Optimised 18 μm CT surface	16,804,493	31,999,434	Surface triangles; approximate pore geometry
Legacy PBR appearance surface	16,272,624	32,687,264	Filled/repaired geometry with old Z calibration
Calibrated L2 mechanical proxy	189,455	963,710	Volume tetrahedra; closed-solid approximation

3 Mechanical model and constitutive audit

3.1 Executed small-strain problem

Let $\Omega \subset \mathbb{R}^3$ denote the calibrated closed-solid proxy. The displacement is $\mathbf{u}$, the linear strain $\varepsilon(\mathbf{u}) = (\nabla\mathbf{u} + \nabla\mathbf{u}^T)/2$, and

$$\boldsymbol{\sigma} = \lambda\,\mathrm{tr}(\varepsilon)\mathbf{I} + 2\mu\varepsilon, \quad \lambda = 29.0\ \mathrm{GPa}, \quad \mu = 28.5\ \mathrm{GPa}. \tag{5}$$

These are assumed homogeneous isotropic constants, not measured fossil properties. They imply

$$\begin{aligned} E &= \mu\frac{3\lambda + 2\mu}{\lambda + \mu} = 71.373913\ \mathrm{GPa}, \\ \nu &= \frac{\lambda}{2(\lambda + \mu)} = 0.252173913, \\ \kappa &= \lambda + \frac{2\mu}{3} = 48.0\ \mathrm{GPa}. \end{aligned} \tag{6}$$

Measurements on extant shark teeth support the relevance of heterogeneous mineralised tissues [11]; they do not establish these exact constants for a fossil with an unknown alteration history.

The model omits body forces. A chosen exterior face patch Γ_P carries inward traction $\mathbf{t} = -\sigma_0\mathbf{n}$, $\sigma_0 = 1$ MPa, while the remaining unconstrained exterior is traction free. Nodal degrees of freedom in two geometric root-cap volumes are fixed to zero. This is a rigid support idealisation: it constrains nodes throughout the selected caps, not solely nodes on an anatomical attachment surface. Write V_h for continuous piecewise-affine vector fields with those nodal constraints. The discrete weak problem is

$$\int_\Omega \boldsymbol{\sigma}(\mathbf{u}_h) : \varepsilon(\mathbf{v}_h)\,\mathrm{d}V = -\int_{\Gamma_P} \sigma_0\mathbf{n}\cdot\mathbf{v}_h\,\mathrm{d}A, \qquad \mathbf{v}_h \in V_h. \tag{7}$$

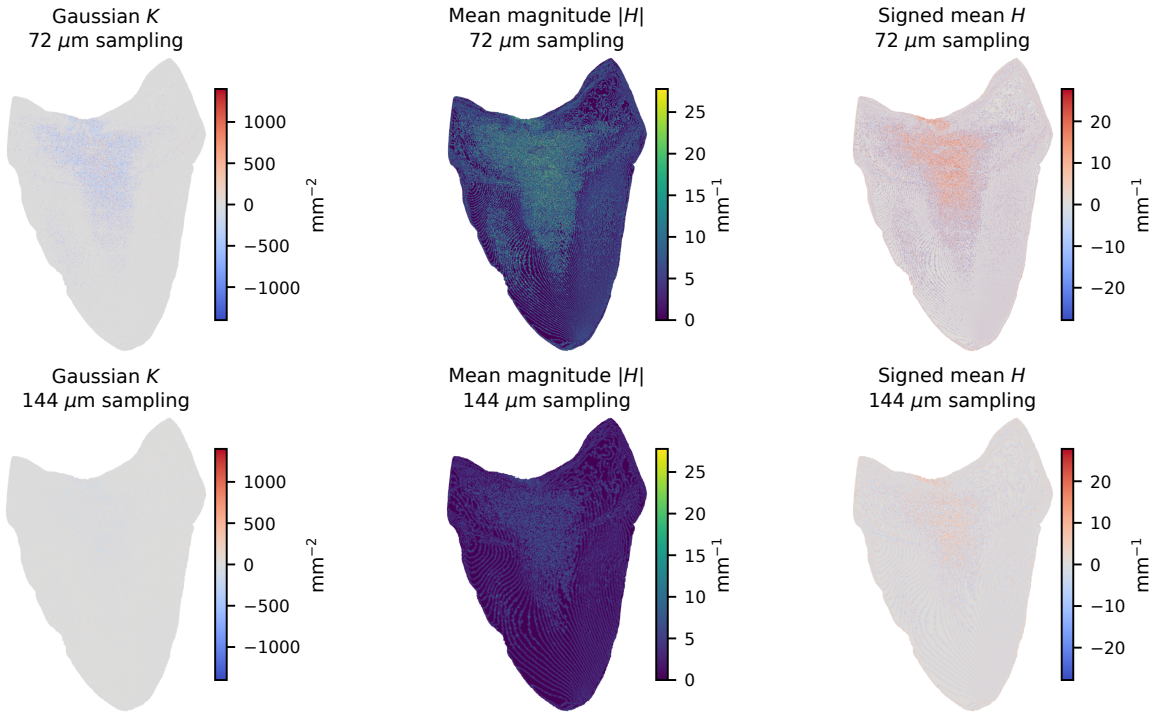


Figure 3: Recomputed curvature from the native saved binary CT mask. Rows use explicit 72 and 144 μm sampling for display; the source mask remains at 18 μm. Columns show area-normalised Gaussian curvature, mean-curvature magnitude and outward-positive signed mean curvature. Each column shares the same numerical colour limits across rows, with pooled 99th-percentile tail clipping. These resampled visualisations expose scale dependence and do not measure native-resolution pore geometry or prove preservation under simplification. Meshes and numerical limits are archived.

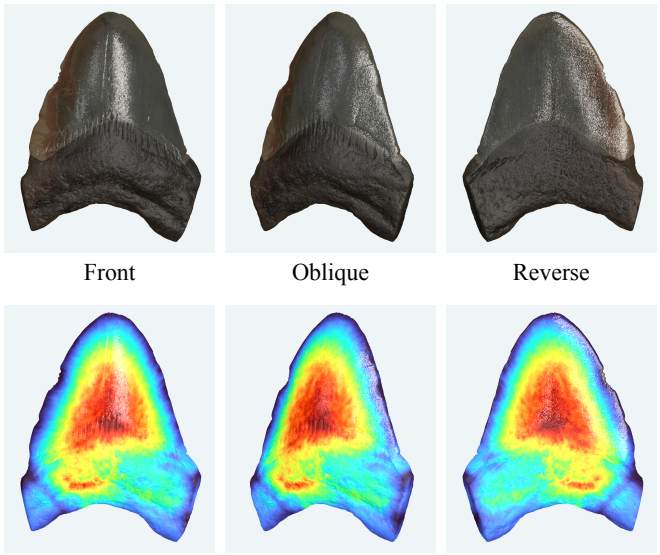


Figure 4: Recovered high-definition appearance exports: natural colour (top) and reference-image false colour (bottom). Both rows are illustrative legacy PBR renders. The bottom row is **not a calculated stress field**; no stress scale, fracture interpretation or spatial validation is assigned to its colours. The unseen reverse colours are extrapolated. Actual numerical stress is shown separately in Figure 5. Crop coordinates and source hashes are preserved.

No nonlinear constitutive model, explicit contact, prey medium, fatigue law or tissue interface is solved in the new pressure experiment.

For each tetrahedron e, an affine displacement gives constant strain and stress. Element stiffness is $K_e = V_e B_e^T C B_e$, with engineering shear strains and the matching isotropic elasticity matrix C. Consistent constant-face traction is shared equally among the triangle's three nodes. The assembled system is reduced to free degrees of freedom, solved, and used to recover element stress without nodal averaging. The reported scalar is

$$\sigma_{\mathrm{vM}} = \sqrt{\tfrac{3}{2}\,\mathrm{dev}\,\boldsymbol{\sigma} : \mathrm{dev}\,\boldsymbol{\sigma}}, \qquad \mathrm{dev}\,\boldsymbol{\sigma} = \boldsymbol{\sigma} - \tfrac{1}{3}\,\mathrm{tr}(\boldsymbol{\sigma})\mathbf{I}. \quad (8)$$

It summarises distortional stress. It is not a validated failure measure for this fossil tooth.

3.2 What tangent matching would require

The archived manuscript compared St. Venant–Kirchhoff (StVK), neo-Hookean, Ogden and Mooney–Rivlin expressions as if their listed parameters gave the same infinitesimal tangent. That assertion depends on the exact strain-energy convention. Let $F = I + \nabla\mathbf{u}$, $J = \det F$, $\bar{I}_1, \bar{I}_2$ be isochoric invariants, and $\bar{\lambda}_i$ isochoric principal stretches. Representative forms are

$$W_{\mathrm{StVK}} = \tfrac{\lambda}{2}(\mathrm{tr}\,E_G)^2 + \mu\,\mathrm{tr}(E_G^2), \quad E_G = \tfrac{1}{2}(F^TF - I), \quad (9)$$

$$W_{\mathrm{NH}} = \tfrac{\mu}{2}(\bar{I}_1 - 3) + \tfrac{\kappa}{2}(J-1)^2, \quad (10)$$

$$W_{\mathrm{Ogden}} = \frac{\mu_1}{\alpha_1}\left(\sum_{i=1}^{3}\bar{\lambda}_i^{\alpha_1} - 3\right) + \tfrac{\kappa}{2}(J-1)^2, \quad (11)$$

$$W_{\mathrm{MR2}} = C_{10}(\bar{I}_1 - 3) + C_{01}(\bar{I}_2 - 3) + \tfrac{\kappa}{2}(J-1)^2, \quad (12)$$

$$W_{\mathrm{MR5}} = W_{\mathrm{MR2}} + C_{20}(\bar{I}_1 - 3)^2 + C_{11}(\bar{I}_1 - 3)(\bar{I}_2 - 3) + C_{02}(\bar{I}_2 - 3)^2. \quad (13)$$

The five-parameter expression has five explicit deviatoric coefficients. For these conventions the infinitesimal shear modulus is μ in StVK/NH, $\mu_1\alpha_1/2$ in Ogden, and $2(C_{10} + C_{01})$ in MR2/MR5. Simple shear directly verifies these coefficients by the quadratic energy term. In particular, at $\alpha_1 = 2$, this Ogden convention has effective shear modulus μ_1, not $\mu_1/2$.

The MR5 value is an archived export setting, whereas the former report called its set unspecified. It is not silently substituted into the new solve. Matching one small-strain tangent would still not establish agreement at finite strain or biological validity. Consequently, equal-looking legacy maxima are not used as evidence that five calibrated tissue laws yield the same result.

3.3 Reference pressure and current pressure

The finite-deformation equations are retained only to resolve the archived notation. If Ω_0 is the reference domain, N its normal, $P = \partial W/\partial F$ and the current Cauchy traction is $\boldsymbol{\sigma}n = -pn$, Nanson's formula gives

$$PN = -pJF^{-T}N. \quad (14)$$

A reference-surface weak form for follower pressure must therefore use $-pJF^{-T}N$ in its external virtual work. The archived $PN = -pn$ and reference-area integral cannot represent current pressure without this transformation. A prescribed reference dead traction is a different, explicitly chosen load. Equation (7) is a small-strain pressure experiment, so its use of the undeformed face normal is consistent with that approximation.

4 Specified numerical experiment and regional results

4.1 Coordinates, physical masks and pressure patch

An exact tetrahedral-volume covariance determines the proxy's principal axes, avoiding vertex-density bias. The longitudinal axis is oriented toward positive source Z (rootward); the thickness axis is oriented toward positive source X. Its positive face is a geometric convention, not an independently verified labial assignment. A transverse axis completes the orthonormal frame. The normalised longitudinal coordinate is $s \in [0, 1]$ from apex to root. At each s, transverse limits are linearly interpolated from 40 longitudinal bins of the L0 exterior vertices, defining local width coordinate r. Thickness coordinate ζ uses the full L0 thickness range about its midpoint.

Table 8 gives the six selection rules. They select L0 tetrahedra by their centroids; each region is then the union of those entire cells. Refined child centroids are located within L0 parents using checked barycentric containment and inherit the parent region. No nearest-cell extrapolation is allowed. Thus refinement leaves each region's physical volume unchanged. The selected regions omit some intervening material and are not an exhaustive anatomical partition.

On the L0 exterior, a face enters the pressure patch if its centroid has $0.18 \le s \le 0.64$, $|r| \le 0.75$ and its outward normal has thickness-axis projection at least 0.5. Child exterior triangles inherit the original patch through checked planar/barycentric containment. Total loaded area remains 483.286722 mm^2 on all three meshes, with identical net pressure force to relative tolerance 10^{-9}. The total scalar pressure-times-area is 483.287 N; the norm of the vector resultant is smaller because the local normals differ.

For each transverse half of the root, its own longitudinal endpoint defines the cap. Fixed nodes satisfy $s \ge 0.80$ and lie within a normalised longitudinal depth 0.04 of that half's endpoint. Both caps constrain all three displacement components. The same geometric rule is evaluated on each mesh; its nodal representation changes during refinement. This is not a measured periodontal attachment. The root-side summary band is not the set of fixed support nodes.

4.2 Meshes and numerical verification

Two conforming refinements of the archived L0 mesh produce L1 and L2, using marked-all-element refinement in scikit-fem [12]. They preserve the same polyhedral domain. Element matrices, tractions and stress recovery are assembled by the accompanying explicit Python implementation. L0 is solved by sparse direct factorisation; L1 and L2 use conjugate gradients with smoothed-aggregation algebraic multigrid and six rigid-motion candidates [13]. L2 has 563,385 free displacement degrees of freedom and required 53 iterations.

Table 4: Constitutive tangent audit of the archived parameter claims. These are algebraic comparisons, not five newly executed nonlinear solves.

Form	Archived entry and implied tangent	Matching requirement for this baseline
StVK	$\lambda = 29$, $\mu = 28.5$ GPa	Defines the assumed linear target
Neo-Hookean	$\mu = 28.5$, $\kappa = 48.6$ GPa	Use $\kappa = 48.0$ GPa for exact target
Ogden, stated convention	$\mu_1 = 57$ GPa, $\alpha_1 = 2$: $\mu_{\text{eff}} = 57$ GPa	$\mu_1 = 28.5$ GPa when $\alpha_1 = 2$
MR2	$C_{10} = 20$, $C_{01} = 8.5$ GPa: $\mu_{\text{eff}} = 57$ GPa	$C_{10} + C_{01} = 14.25$ GPa
MR5 archived export	$C_{10} = 12$, $C_{01} = 3$ MPa: $\mu_{\text{eff}} = 30$ MPa	Export is not matched to 28.5 GPa; higher coefficients do not change the infinitesimal shear tangent

Table 5: Status of earlier stress evidence after the load and source audit.

Item	What is recoverable	Permitted use
Original regional table	Printed $Q_{0.95}$ values: 0.89, 21.8, 20.9, 46.2, 14.7, 0.47 MPa	Historical claims only; original anatomical masks and 1 MPa field unavailable
30 October recalculation	Closed-solid P1 solve on the old Z scale; equal-element quantiles	Reproducible earlier experiment; superseded, not numerically identical to this table
Legacy frontal field figure	Original 2,444×3,016-pixel export; colour scale in Pa	Different archived field; not evidence for a 1 MPa input pressure
Legacy PBR false colour	Original appearance exports and explicit no-field metadata	Illustrative image only; no stress values recovered from colour
New regional pressure field	Full elements, displacement, stresses, volumes, masks, face IDs and support IDs	Direct numerical evidence for the defined proxy experiment

Mesh quality is reported using $q_e = 12(3V_e)^{2/3}/\sum_{i<j}\ell_{ij}^2$, equal to one for a regular tetrahedron, together with positive-volume checks. Full minima, first percentiles and medians are archived rather than describing refinement as automatically improving every element. The domain is millimetre-scale and the finest maximum edge is 2.473 mm. The calculation does not establish a prescribed number of elements across each individual serration notch.

The implementation checks are necessary consistency tests. They do not validate the biological support law or compensate for segmentation bias. A very small algebraic residual also does not establish mesh convergence of a reported stress functional. Conversely, increasing pointwise maxima under refinement need not invalidate region integrals or quantiles; those outputs must be tested separately.

4.3 Volume-weighted stress quantiles

For a region R of positive volume, the physical summary is the inverse volume-distribution quantile

$$F_R(t) = \frac{1}{|R|}\int_R \mathbf{1}_{\{\sigma_{\text{vM}}(\mathbf{x})\le t\}}\,\mathrm{d}V, \qquad Q_q(R) = \inf\{t : F_R(t) \ge q\}. \tag{15}$$

For constant P1 element stress this is implemented exactly as a weighted empirical CDF:

$$F_{R,h}(t) = \frac{\sum_{e\subset R} V_e \mathbf{1}_{\{\sigma_{\text{vM}}^{(e)}\le t\}}}{\sum_{e\subset R} V_e}. \tag{16}$$

The archived rows report $q = 0.90, 0.95, 0.99$. Independent CDF inequalities immediately below and at all 54 reported values verify the readback across the three meshes. Equal-element type-7 sample quantiles are also saved for comparison, but are not the primary regional physical measure: refinement changes the number and distribution of elements. For example, on L2 the central $Q_{0.95}$ is 10.555 MPa by volume weighting and 13.112 MPa by equal-element sampling. The convention must therefore accompany the number.

Define $\text{SAF}_q(R) = Q_q(R)/\sigma_0$. Because $\sigma_0 = 1$ MPa, Table 9's SAF numbers equal its stresses expressed in MPa, although SAF is dimensionless. SAF is a pressure-normalised regional stress factor. It is distinct from the edge-to-central-reference ratios, which are 2.803 for R_A and 1.381 for R_B at $q = 0.95$.

The baseline regional ordering is

$$R_{\text{root}} > R_{\text{cr}} > R_A > R_B > R_0 > R_{\text{tip}} \quad \text{by } Q_{0.95}. \tag{17}$$

The small value at the macroscopic apex follows from this pressure patch and support configuration; it does not test a load applied directly to individual serration tips. The high root-side value is attachment-sensitive and cannot be read as proof that fossil roots commonly failed before crowns. The former tip-dominant, low-stress-root table is not supported by this replacement experiment.

4.4 Refinement behaviour and limits of percentile robustness

Figure 6 shows the actual three-mesh regional comparison. The last-step relative changes in $Q_{0.95}$ are 1.80, 0.64, 0.90, 5.89, 1.85 and 0.17% for $R_0, R_A, R_B, R_{\text{tip}}, R_{\text{cr}}, R_{\text{root}}$. A 2% descriptive threshold is met by five regions; even a 5% threshold is not met by the macroscopic apex. This is evidence of limited finite-mesh stability, not an asymptotic convergence proof. $Q_{0.90}$ and $Q_{0.99}$ and weighted exceedance-set overlaps are archived separately; stability of one quantile cannot certify them all.

The maximum element stress increases from 875.0 to 1,236.8 to 1,914.8 MPa and is treated as a diagnostic near the idealised supports. It is not converted to a finite strength or identified as a demonstrated serration singularity. Angular-corner theory depends on angle and boundary conditions [14]; it does not supply one universal singular exponent for every tooth notch, especially when physical rounding and imaging resolution matter.

A percentile is not automatically immune to a single divergent sample. In a type-7 sample of 50 values, with 49 zeros and one value M, $Q_{0.99} = 0.51M$. For physical volume quantiles a useful boundedness statement instead requires a norm bound:

Proposition 1. *If $0 < |R| < \infty$ and $\sigma_{\text{vM}} \in L^2(R)$, then, for $0 < q < 1$,*

$$Q_q(R) \le \frac{\|\sigma_{\text{vM}}\|_{L^2(R)}}{\sqrt{(1-q)|R|}}. \tag{18}$$

Proof. The fraction of R on which $\sigma_{\text{vM}} > t$ is at most $\|\sigma_{\text{vM}}\|_{L^2(R)}^2/(t^2|R|)$ by Markov's inequality. Choose t at the

Table 6: Executed nested volume meshes on the same calibrated closed-solid proxy domain.

Level	Nodes	Tetrahedra	Free DOF	$h_{\max}$	$\bar{h}_V$	Peak σ_{vM}
L0	13,584	55,852	40,224	4.447	1.946	875.0
L1	55,315	271,300	164,361	3.102	1.302	1236.8
L2	189,455	963,710	563,385	2.473	0.854	1914.8

Element edge lengths are in mm; stresses in MPa. $\bar{h}_V$ is the volume-weighted mean of each tetrahedron's longest edge. Surface-face counts in Table 3 are different objects from these volume-element counts.

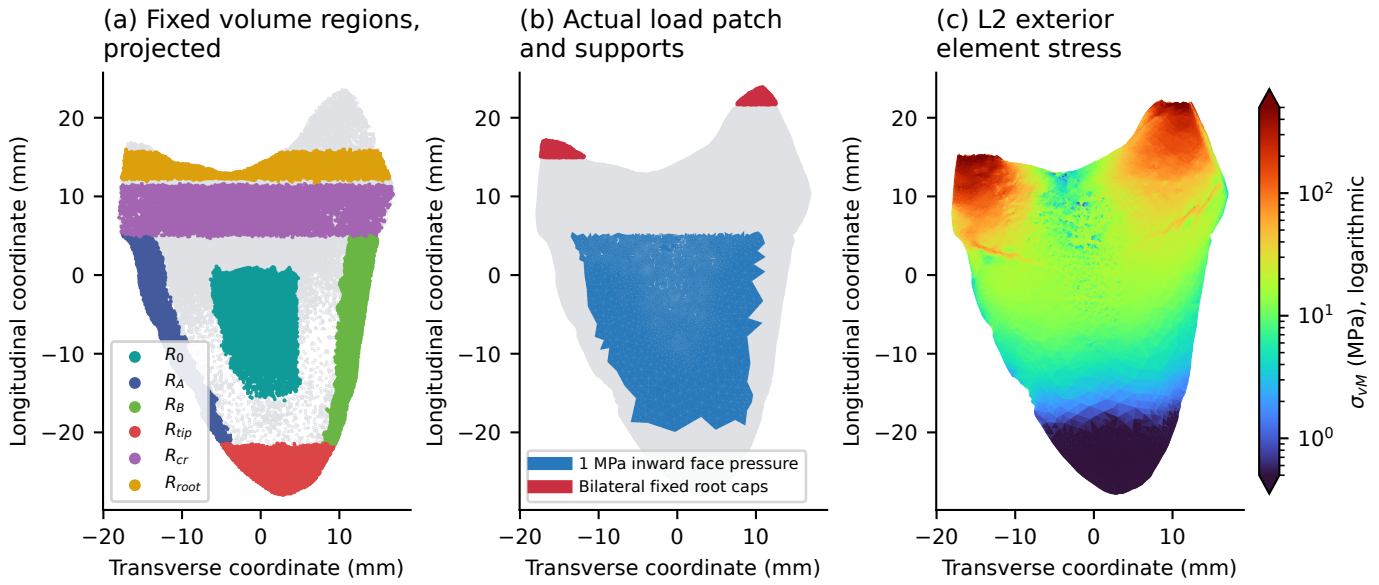


Figure 5: Recomputed mask/load/stress plate replacing the earlier screenshot. (a) Projections of the six fixed volume-region unions; interior points are projected for visibility, not labelled as exterior anatomy. Plotting is subsampled only in this panel; calculations use every selected element. (b) The actual pressure face patch (blue) and fixed root-cap nodes (red). (c) L2 exterior triangles coloured by their owning tetrahedron's unsmoothed von Mises stress. The logarithmic display range is 0.5–500 MPa, with out-of-range values saturated; the numerical maximum is 1,914.835 MPa. No displacement amplification, PBR reference colour or inferred stress interpolation is used.

Table 7: Executed implementation and equilibrium checks; these verify the numerical problem, not fossil behaviour.

Check	L2 result
Affine traction patch error	3.9627×10^{-15}
Rigid-motion strain (absolute)	3.4694×10^{-18}
Free-system residual	2.6715×10^{-10}
Net force imbalance	4.4666×10^{-12}
Net moment imbalance	2.3218×10^{-12}
Work–energy identity error	5.3538×10^{-12}

The affine and rigid-motion tests use an independent five-tetrahedron cube. The other quantities are dimensionless relative diagnostics, normalised as specified in the archived solver. All 16 sensitivity cases also passed their 2×10^{-7} equilibrium tolerance; their largest residual was 4.43×10^{-9}.

displayed bound, for which $F_R(t) \geq q$. The definition of the inverse quantile gives the conclusion. □

A mesh-uniform bound requires mesh-uniform L^2 control. This bound alone supplies neither quantile convergence nor stability of a specific high-stress location. The new table's validation therefore relies on direct field readback and refinement comparisons, not a blanket robustness assertion.

5 Sensitivity calculations and procedural evidence checks

5.1 Executed sensitivity set

Sixteen specified L0 cases vary the support, load direction/footprint and elastic constants. Nine cases combine bilateral fixed caps, one-lobe fixed caps, or a three-direction elastic foundation with traction directions $-30°, 0°, +30°$. The local tangential direction is the normalised projection of the transverse axis onto each loaded face. Oblique cases retain a 1 MPa resultant traction magnitude; their normal component is $\cos 30°$ MPa, rather than 1 MPa. Positive/negative angles are signed by the recorded transverse convention.

The foundation applies nodal springs to exterior faces whose vertices all belong to the root caps, with tributary-area weighting and assumed stiffness 10^{11} N/m^3 in all three directions. This value is a numerical sensitivity parameter, not a measured periodontal stiffness. Partial support fixes only the positive-transverse root lobe. The other seven cases use bilateral normal loading: narrow and broad face footprints; $\nu = 0.20$ and 0.30 at fixed baseline E; cap depths 0.03 and 0.05; and doubled E at fixed ν. The narrow patch additionally requires $|r| \leq 0.45$; the broad patch uses $0.12 \leq s \leq 0.70$, $|r| \leq 0.90$ and the same normal-projection test. All exact face IDs/rules and case parameters are archived.

The full baseline ordering occurs in 13 cases. In all three one-lobe support cases it becomes $R_{\text{root}} > R_{\text{cr}} > R_B > R_0 > R_A > R_{\text{tip}}$. Thus both edge sides are not invariably above the central reference, and the relative A/B contrast is not support-invariant. The root-side and transition regions remain the two highest and the macroscopic apex the lowest of these six summaries in all 16 tested cases. This statement is restricted to the specified case set and geometry; it does not establish a universal tooth stress hierarchy.

5.2 Material and pressure scaling checks

For homogeneous linear elasticity at fixed ν, traction and geometric Dirichlet constraints, scaling E by c scales K by c, so displacement scales by $1/c$ and stress is unchanged. Scaling pressure by a scales displacement and stress by a. These identities fail to settle questions involving heterogeneous stiffness ratios, nonlinear geometry, contact, damage or stiffness-dependent attachment.

Six additional L0 solves combine $E = 40, 90$ GPa with $p = 0.5, 1, 5$ MPa and directly check those scalings against the recovered full field. They are algebraic verification within the stated linear model, not new evidence of nonlinear material agreement. Varying ν changes the stiffness ratios and is therefore an actual stress sensitivity, represented in the 16-case set. No nonlinear material law or material interface was calibrated or re-solved for this revision.

5.3 Table 11: performed checks and remaining obligations

The former Table 11 separated a biomechanics manuscript from a palaeobiological interpretation and was followed by an unexecuted roadmap. Table 11 replaces that roadmap with the evidence actually obtained. "Partial" means that a finite numerical check exists but does not satisfy the full scientific obligation. In particular, success in residual checks is not experimental validation.

This audit permits a concrete conclusion about the simulated experiment while retaining explicit open biological obligations. Neither route is validated merely by adding illustrations. The paper therefore ends with the executed study's findings rather than another proposed workflow or a calendar of future tasks.

Table 8: Frozen physical region unions and their L2 populations. Regions do not partition the entire tooth.

Region	L0 centroid-selection rule	L2 cells	Volume (mm^3)
R_0	$0.25 \le s \le 0.55,\ \lvert r\rvert \le 0.35,\ \lvert\zeta\rvert \le 0.5$	31,130	720.574
R_A	$0.12 < s < 0.64,\ r \le -0.78$	25,905	171.580
R_B	$0.12 < s < 0.64,\ r \ge 0.78$	16,324	143.799
R_{tip}	$s \le 0.12$	13,065	152.952
R_{cr}	$0.64 \le s \le 0.76$	397,306	1367.885
R_{root}	$0.78 \le s \le 0.84$	71,316	316.788

s runs from the macroscopic apex (0) to the root (1); r is locally normalised transverse position and ζ normalised thickness. R_A/R_B are geometric sides, not verified mesial/distal identities. R_{tip} is the macroscopic apex, not individual serration tips. R_{cr} is a geometric transition band, not a segmented EDJ.

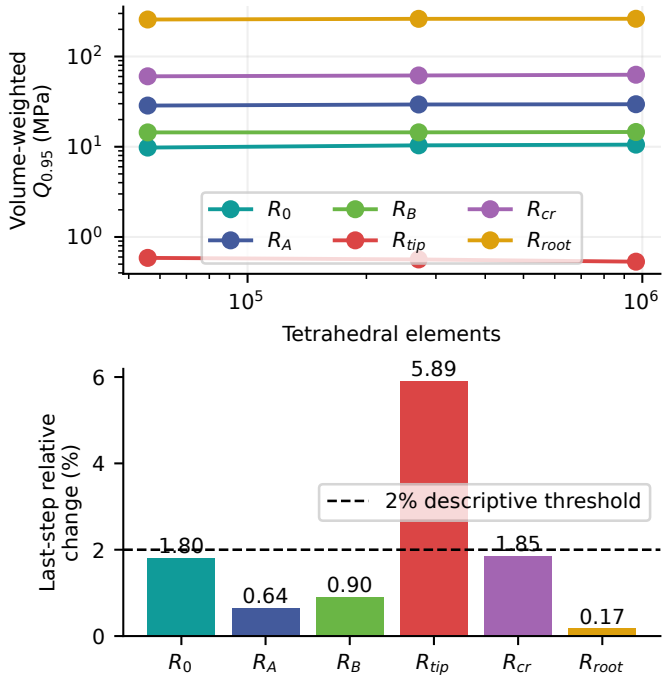


Figure 6: Executed refinement study on a fixed proxy domain, pressure patch and physical region unions. Left: regional $Q_{0.95}$ versus volume-element count; both axes are logarithmic. Right: relative L1–L2 changes, using L2 as denominator. The dashed 2% line is a descriptive acceptance threshold, not a theorem. The macroscopic apex remains sensitive to refinement.

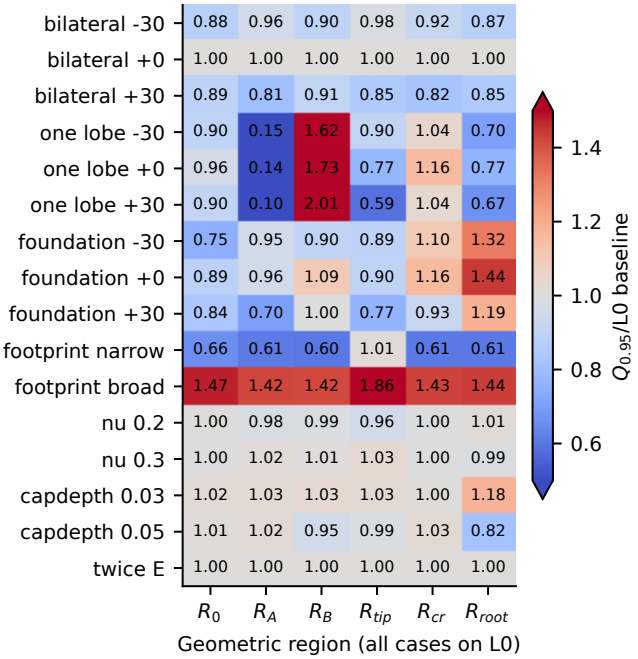


Figure 7: Sixteen actual L0 sensitivity solves. Each cell is the regional $Q_{0.95}$ divided by its bilateral normal-pressure L0 baseline, so 1 means no change. Colours saturate outside 0.5–1.5 while printed values retain the numerical ratio. The matrix is a finite set of declared assumptions; it is not a probabilistic uncertainty distribution and has not been repeated on L2.

Table 9: Region-wise percentile stress and stress-amplification factors under a standardised crown-face pressure $\sigma_0 = 1$ MPa. Values are region summaries of σ_{vM} (MPa), not pointwise fracture criteria.

Region	$Q_{0.90}$	$Q_{0.95}$	$Q_{0.99}$	$\text{SAF}_{0.95}$	L1–L2 change
R_0	9.057	10.555	13.084	10.555	1.80%
R_A	24.407	29.587	48.052	29.587	0.64%
R_B	12.227	14.572	18.507	14.572	0.90%
R_{tip}	0.446	0.533	0.716	0.533	5.89%
R_{cr}	44.705	62.811	112.680	62.811	1.85%
R_{root}	188.245	261.966	389.605	261.966	0.17%

Primary quantiles are inverse volume-weighted empirical CDFs of the piecewise-constant P1 element stress on L2 (963,710 tetrahedra). $\text{SAF}_{0.95} = Q_{0.95}/\sigma_0$ is dimensionless. Last-step changes are $|Q^{(2)}_{0.95} - Q^{(1)}_{0.95}|/Q^{(2)}_{0.95}$. The apex changes by 5.89%; the other five regions change by less than 2%. This table supersedes the original and the 30 October single-mesh tables; geometry calibration, physical mask definition, weighting and refinement have all changed.

Table 10: Finite sensitivity envelope of regional $Q_{0.95}$ on L0 (16 specified cases).

Region	Minimum (MPa)	Baseline (MPa)	Maximum (MPa)
R_0	6.459	9.790	14.421
R_A	2.894	28.602	40.665
R_B	8.715	14.419	28.967
R_{tip}	0.345	0.587	1.091
R_{cr}	36.763	60.347	86.047
R_{root}	156.449	257.059	371.202

This is a deterministic range across the specified cases, not a confidence interval or exhaustive uncertainty envelope. In the nine direction/support cases the resultant traction magnitude is 1 MPa; at $\pm 30°$ the normal component is $\cos 30°$ MPa. Six additional pressure/stiffness scaling solves are archived separately.

6 Discussion

6.1 Regional localisation and attachment sensitivity

The central result is conditional but determinate: on the specified closed-solid proxy under distributed face pressure and bilateral fixed caps, both edge-side $Q_{0.95}$ values exceed the central reference, whereas the root-side and transition bands are higher and the macroscopic apex lower. These outputs differ substantially from the original printed table. The change cannot be attributed solely to mesh refinement: the original anatomical experiment was unavailable, the old Z scale was corrected, masks were fixed as physical unions and volume weighting replaced element-count weighting.

The one-lobe cases demonstrate that support assumptions can reverse the A/B relation and put one side below the central reference. A regional hierarchy should therefore be reported together with its load and support definitions. The study does not establish stress concentration at individually identified serration tips, and it cannot identify the high-root stress in this idealisation as a fossil failure location. Domain fidelity is a separate uncertainty: the native CT internal voids and the mineralised tissue architecture are absent from the mechanical proxy. Refining that proxy only reduces discretisation uncertainty within it.

The calibrated L2 maximum displacement is 0.619 mm. This value and the reported stress levels provide diagnostics of the imposed load and stiffness assumptions, not an independent check of finite-strain accuracy. No geometrically nonlinear comparison has been performed. Linear proportionality to pressure should therefore be read as a property of the model over the

Table 11: Procedural evidence audit corresponding to the former Table 11 and its seven review modules.

Obligation	Executed evidence	Status and boundary
CT metadata	All 3,050 archived headers agree on 18 μm spacing; three fresh raw-header spot checks agree. Source Z override identified.	Completed for available geometry metadata; absolute scanner calibration unverified.
Segmentation sensitivity	Native/legacy/optimised geometry and seven saved CT sections compared; no threshold $\pm 5\%$ full-volume rerun.	Incomplete; threshold uncertainty and repeatability remain unquantified.
Topology and geometry	Archived exhaustive edge/link checks; native/optimised volumes and sampled distances; fresh curvature sphere/scale benchmarks.	Partial; no exhaustive self-intersection test or anatomical pore validation.
Mesh convergence	Three nested meshes; all 54 regional quantiles; quality statistics; volume-weighted L1–L2 95%-exceedance Dice overlaps.	Partial; apex $Q_{0.95}$ changes 5.89%; notch resolution and asymptotic convergence not established.
Boundary conditions	Three supports by three directions, two additional footprints and two cap-depth changes.	Partial; finite L0 set, no pure lateral case or complete factorial study, no measured attachment.
Material uncertainty	$\nu = 0.20, 0.30$; doubled-E check; six $E = 40, 90$ GPa and $p = 0.5, 1, 5$ MPa scaling solves; corrected hyperelastic tangents.	Partial; homogeneous linear sensitivity only; no calibrated nonlinear/fossil tissue law.
Regional extraction	Full element field, volumes, physical masks and inverse-CDF extraction; all Table 9 values checked by independent CDF inequalities.	Completed for the new geometric experiment; original anatomical result remains unreproducible.
Numerical verification	Affine traction patch, rigid motions, force/moment balance, work–energy identity, residuals and scaling.	Completed within the declared tests; not physical validation.
Fracture screening	Stress-summary limitations and required crack/interface inputs stated; former numerical fracture pressure removed.	No validated strength, toughness, flaw population or fracture threshold.
Biological validation	Original two photographs and comparative primary literature reviewed; no new specimen survey or damage map.	Incomplete; dental position, tissue boundaries and fossil damage statistics unresolved.
Reproducibility	Editable manuscript, scripts, meshes, raw fields, masks, CSV tables, image sources/crop records and hashes supplied.	Completed for deposited new calculations; large native CT sources identified separately.
Manuscript route	New evidence supports a conditional computational tooth-model study.	Palaeobiological, feeding and ecological mechanisms are not established.

algebraically tested range, not as proof that real tissue remains linearly elastic up to 5 MPa applied pressure.

6.2 Fracture, interfaces and geometric scaling

A von Mises percentile cannot identify brittle crack initiation in an anisotropic or heterogeneous dental tissue. Even an elementary mode-I estimate $K_I = Y\sigma_n\sqrt{\pi a}$ requires crack-normal stress σ_n, flaw size a, geometry factor Y and a relevant toughness. It cannot be evaluated by silently replacing σ_n with a regional von Mises quantile. The former illustrative critical pressure depended on unmeasured fossil toughness and a chosen flaw length; no numerical critical pressure is retained here.

Interface crack deflection depends on elastic mismatch, energy-release-rate competition and interface/bulk toughness [15]. A continuous single-material solid with no identified interface cannot test enameloid–dentine spalling. The texture boundary in a colour render supplies neither an EDJ segmentation nor those constitutive inputs. Likewise, photographs alone do not distinguish feeding damage from taphonomic alteration, collection damage or preparation.

A simple rectangular-beam analogy can isolate one geometric effect without predicting behaviour. For width W, bending thickness t, lever arm L and an assumed allowable tensile stress σ_a, elementary bending gives $F_a = \sigma_a W t^2/(6L)$. If $t = \tau H$ and $L \approx H$, then $F_a \approx \sigma_a W \tau^2 H/6$. Thinner shapes have lower capacity under the matched assumptions, but width, height, lever arm, loading direction and material must also be matched. This scaling does not establish a universal fractional performance deficit or a prey-size ceiling.

6.3 Functional interpretation and comparative limits

Dynamic cutting experiments, structural tooth models and extant tissue measurements answer complementary questions [1, 2, 11]. A fossil functional argument would additionally require reliable dental orientation, comparable geometry/load cases, a damage-location survey and independent material/attachment constraints. None is supplied by a false-colour render. The present photographs and computed regional fields do not show that tooth damage was beneficial, that replacement caused a particular predatory strategy or that an interface was an adapted failure plane.

The available evidence supports a reproducible stress-localisation experiment and identifies several failures of the archived numerical narrative. It does not estimate a whole-animal bite force, maximum body length, prey preference, competitive superiority, niche partition or extinction mechanism. Removing those inferences is a consequence of what was measured and solved, rather than a claim that they are scientifically unanswerable by other evidence. The decisive endpoint for this paper is the verified conditional regional comparison and its demonstrated sensitivity.

Data, code and reconstruction record

All text, equations and tables in this revised manuscript are editable LaTeX. The exact typesetting source of the supplied 26-page PDF was unavailable; this is a scientifically revised reconstruction, not a claim to recover identical original source bytes. The older recovered `pde_paths/debunk_myth.tex` and the previous PDF are retained unchanged in the source archive. Numerical meshes, full element fields, physical mask IDs, CSV tables, checks, source hashes and executable scripts accompany this version. The native 18 μm segmentation and the two large 3MF surfaces remain in the project's `full_3D` directory and are identified by hash in the manifest. Native CT pores are not resolved in the stress solve.

The archival native reconstruction reports were inspected; their all-header consistency checks were supplemented by fresh raw-header reads at slices 0, 1525 and 3049. The three stress meshes, 16 sensitivity cases, six scaling cases and curvature benchmarks were executed for this revision. The audit is a sequential computational and editorial self-review, not independent peer review. Image provenance and remaining scientific obligations are recorded in the accompanying claim ledger. Photograph originals were extracted losslessly from the supplied report; the third introductory panel is a native-resolution frame from the recovered animation.

Authorship record

The original report credits Eliguli Han for the computational project and manuscript and lists Leslie Greengard as faculty advisor. Those attributions are retained as project history. The reconstructed calculations and editorial changes do not imply an additional advisor endorsement or an independently validated biological material calibration.

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7 Conclusions

The archived CT geometry is sampled at 18 μm in all three directions. The old 16 μm Z override and the manuscript's 50/50/100 μm entry are inconsistent with that record. Native CT surfaces, legacy appearance models and the closed-solid stress proxy must consequently remain distinct.

The recalibrated pressure experiment produces a reproducible replacement Table 9. On the finest executed mesh, the central, A-side and B-side $Q_{0.95}$ stresses are 10.555, 29.587 and 14.572 MPa; the macroscopic apex, transition and root-side values are 0.533, 62.811 and 261.966 MPa. Five regions show less than 2% last-step change, while the apex remains at 5.89%. Implementation checks support these finite-mesh numerical outputs; they do not certify anatomical or constitutive accuracy.

The baseline model has edge-to-reference factors 2.803 and 1.381. Three of the 16 sensitivity cases reverse the A/B ordering and put A below the central reference. The regional contrast is therefore support-dependent. In the tested set, root-side and transition summaries remain highest and the macroscopic apex lowest. The original tip-dominant, low-stress-root interpretation is not supported by the new experiment.

The scientific conclusion is a conditional localisation result for a specified single-material tooth proxy. Individual serration failure, fossil fracture pressure and organism-level feeding or ecological mechanisms have not been demonstrated. Authentic photographs, recovered appearance exports and newly computed stress/curvature figures now have explicit, separate evidentiary roles, and the full editable source and numerical record permit the reported calculation to be checked.

Full-resolution CT source downloads

32M reconstruction - 31,999,434 faces - 528 MB

Megalodon_FullCT_Internal_32M_18um.3mf

Unreduced native - 74,293,846 faces - 753 MB

Megalodon_FullCT_Native_Internal_74M_18um.3mf

Click either source name to download the unchanged 3MF. Geometry retains CT-derived internal voids and DICOM-declared 18 micrometre spacing.

Interactive COMSOL and natural-color 3D archive